EE6406
Analytic and Ensemble Machine Learning
(Lecture 8: Performance Evaluation and Statistical Inference)

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Lecture 8

Performance Evaluation and Statistical Inference

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Introduction

We have learned the main components of learning:

- learning model
- learning loss and cost functions
- optimization procedure

In this lecture, several existing methods for performance evaluation and statistical inference are presented.
Essentially, performance evaluation provides a measure upon how good can a learned predictor or classifier perform and statistical inference provides the confidence of such performance on unseen data.

Empirical Evaluation

Apart from their use in classifier learning, the learning score functions discussed in Lecture 4 can be adopted here for classifier performance evaluation.

- Essentially, the evaluation has to take into consideration how well can the predictor or classifier perform for unseen data.
- Without knowing the unseen data, the evaluation can be performed by holding out some data from the given samples for testing.
- The testing performance shall make use of those measures discussed in Lecture 4 for scoring purpose.

Empirical Evaluation

Hold-Out Test

- Consider a database consisting of M samples $(\boldsymbol{x}_i, y_i)$, $i = 1, 2, ..., M$, where $\boldsymbol{x}_i \in \mathbb{R}^d$ denotes the ith feature sample, and $y_i \in \{0, 1\}$ denotes the corresponding category indicator or target label.
- A *hold-out* evaluation splits the database into two sets namely a training set which consists of m samples and a test set which consists of n samples (i.e., $m + n = M$). The training set is used for classifier learning and the test set is used for evaluation of the learned classifier's performance.
- If $m \approx n$, then a two-fold test can be performed. This means that after the first round of training and testing, the training set and the test set can be swapped and the training and testing processes can be repeated. The two test results are then averaged to produce an overall test result.

Empirical Evaluation

Cross-Validation Test

- It is argued that the hold-out test does not make use of all given data for classifier evaluation. Although the two-fold test serves the purpose of testing all data, it may not be statistically informative to describe the classifier performance particularly for small data sets.
- Suppose we have a small set of M samples for training and testing. Without re-sampling, the largest possible number of test cases without under-training the system is given by the *leave-one-out cross-validation* (LOOCV) method. In this method, one piece of data is kept for testing while the remaining data is used for training. This gives rise to M test results, which are often reported in averaged form.

Empirical Evaluation

Cross-Validation Test

- However, if M is huge, then the leave-one-out test may become impractical since it will take up huge computing effort.
- A practical way to produce the desired statistics without looping the training process for full M times is to perform an *N-fold cross-validation*. Here, the entire data set is partitioned into N equal portions where the positive and negative classes are evenly distributed among these partitions.
- Typical values for N include 2, 5 and 10. The N test results are reported in average form.

Null Hypothesis Significance Test (NHST)

Statistical Significance Test

Statistical inference through hypothesis testing is a method of confirmatory analysis based on data.

- It is often used to decide whether experimental results contain enough information to challenge conventional wisdom.
- In other words, hypothesis testing in statistical inference assesses the confidence in the occurrence of an experimental observation using statistical methods.

Null Hypothesis Significance Test (NHST)

- In empirical studies, it is often desired to conduct experiments to compare between two or among several algorithms on one or more data sets.
- The number of data sets to be used depends much on the data availability in the field of research.
- In the following, common scenarios of comparison such as comparing two classifiers on a single data set, comparing two classifiers over multiple data sets, and comparing multiple classifiers over multiple data sets are covered.

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers on a Single Data Set

McNemar's test:

- This is a non-parametric method applied to nominal data.
- It is available in R, Python and other statistical packages.

Null Hypothesis Significance Test (NHST)

McNemar's test (cont'd):

- The test uses a 2×2 contingency table (Table 1) based on the outcomes of two tests on a sample of n subjects.
- Under the null-hypothesis (H_0) that the probability of occurrences $p_b = p_c$, the two classifiers have no significant difference in their marginal proportions.
  - The numbers b and c are called *discordances* since they refer to the 'No' occurrence for one classifier, but 'Yes' occurrence for the other classifier. A large value of the test statistic gives a small p value and this supports H_1 to reject the null-hypothesis H_0.

Table 1: Contingency table for McNemar's test

	(Classifier B)	
(Classifier A)	'Yes'	'No'
'Yes'	a	b
'No'	c	d

Null Hypothesis Significance Test (NHST)

McNemar's test (cont'd):

The main assumptions made in McNemar's test include:

(i) The classifier pair (A, B) is mutually independent,

(ii) Each classifier A and classifier B can be assigned to one of two possible categories.

Null Hypothesis Significance Test (NHST)

McNemar's test (cont'd):
Comparing Two Classifiers on a Single Data Set

Example 8.1 (Python)

Python:

```
from statsmodels.stats.contingency_tables import mcnemar
table = [[15, 4],
         [25, 6]]
result = mcnemar(table, exact=False, correction=True)
print(f"Statistic:  result.statistic:.4f")
print(f"P-value:  result.pvalue:.4f")
alpha = 0.05
if result.pvalue < alpha:
     print("Reject the null hypothesis:  Significant difference exists.")
else:
     print("Fail to reject the null hypothesis:  No significant difference.")
```

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers over Multiple Data Sets

Averaging over data sets:

- A straightforward approach to compare the difference between two classifiers over multiple data sets is to compute their average classification accuracies across the tested data sets.
- For example, a comparison of average accuracies of different classifiers on a set of related problems such as medical databases for a certain diagnosis from different institutions could be useful.

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers over Multiple Data Sets

Averaging over data sets (cont'd):

- However, it is argued that averaging accuracies across domains could be meaningless if the results on different data sets are not comparable.
- Moreover, averages are susceptible to outliers since one excellent performance on one data set can prevail the overall poor performance, or vice versa.

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers over Multiple Data Sets

***t*-Test**:

- Another way to compare the difference between two classifiers over several data sets is to compute a *t-test* when the data is normally distributed.
- It is available in Matlab (statistics toolbox) and in Python (scipy.stats).
- The testing includes two cases of considerations namely, *unpaired* and *paired* t-tests.

Ref [1]: Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research, 7:1–30.

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers over Multiple Data Sets

t-Test (cont'd):

- An *unpaired* (or independent samples, also known as Student's) t-test is a two-sample location test of the null hypothesis which takes the means of two normally distributed populations being equal.
  - For example, the effect of a treatment has been measured on one population before treatment and on another population after treatment.
- A *paired* (or dependent samples, or repeated measures) t-test however, assumes that the two responses measured are on the same statistical unit.
  - For example, in cancer treatment, an evaluation of effectiveness is to measure the size of tumor before and after the treatment for the same population of patients.

Ref [1]: Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research, 7:1–30.

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers over Multiple Data Sets

***t*-Test (cont'd)**:
When comparing the means of two independent samples, the following assumptions should be met:

(i) Normal distribution of compared populations,

(ii) Same variance for the two compared populations being compared (as though the distribution for one group was merely a shifted of the other group, without changing shape),

(iii) Data being sampled independently for the two compared populations.

Ref [1]: Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research, 7:1–30.

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers over Multiple Data Sets

Wilcoxon Signed-Ranks Test:

- The Wilcoxon signed-ranks test serves as a non-parametric alternative to the paired t-test.
- Consider N data sets, and let d_i represent the difference in accuracies between two classifiers for each set $i = 1, ..., N$.
  - These differences are ranked in ascending order, with the better-performed one having a smaller rank number.
  - When two classifiers have the same accuracy, an average rank is assigned. We denote R^+ as the sum of ranks for data sets where the second algorithm outperformed the first, and R^- as the sum of ranks for the opposite case.
  - In the event of ties (i.e., when $d_i = 0$), the ranks are evenly distributed among the sums.
  - If there is an odd number of ties, one rank is disregarded.

Ref [1]: Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research, 7:1–30.

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers over Multiple Data Sets

Wilcoxon Signed-Ranks Test (cont'd):

- Mathematically, R^+ and R^- can be represented as:

$$R^+ = \sum_{d_i>0} \text{rank}(d_i) + \frac{1}{2}\sum_{d_i=0} \text{rank}(d_i) \quad (1)$$

$$R^- = \sum_{d_i<0} \text{rank}(d_i) + \frac{1}{2}\sum_{d_i=0} \text{rank}(d_i) \quad (2)$$

When $\min(R^+, R^-)$ is smaller than or equal to the critical value in the statistical table, the null hypothesis is rejected.

Ref [1]: Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research, 7:1–30.

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers over Multiple Data Sets

Wilcoxon Signed-Ranks Test (cont'd):

Example 8.2

Table 2: Classification accuracies of LSE-based and TER-based learning

No.	Data	LSE-based Ave(x% accuracy)	TER-based Ave(y% accuracy)	Diff (%) ($d = y - x$)	Rank
1	Shuttle-l-contr	99.03	99.06	+0.03	7
2	BUPA-liver	72.17	72.16	−0.01	3.5
3	Monk-1	89.90	90.55	+0.65	15
4	Monk-2	66.16	68.27	+2.11	16
5	Monk-3	90.39	91.02	+0.62	13.5
6	Pima-diabetes	77.73	77.73	0.00	1.5
7	Tic-tac-toe	98.22	98.24	+0.02	5.5
8	Breast-cancer-W	97.26	97.37	+0.11	8.5
9	StatLog-heart	83.25	83.07	−0.19	11
10	Credit-app	86.65	86.64	−0.01	3.5
11	Votes	94.24	94.27	+0.02	5.5
12	Mushroom	100.00	100.00	0.00	1.5
13	Wdbc	96.18	96.80	+0.62	13.5
14	Wpbc	79.60	79.49	−0.11	8.5
15	Ionosphere	89.18	89.63	+0.45	12
16	Sonar	81.78	81.96	+0.17	10

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers over Multiple Data Sets

Wilcoxon Signed-Ranks Test (cont'd):

Example 8.2 (cont'd)

- The sum of ranks for positive differences between the performances of TER-based learning and LSE-based learning is
  $R^+ = (7 + 15 + 16 + 13.5 + 5.5 + 8.5 + 5.5 + 13.5 + 12 + 10) + (1.5) = 108$.
  On the other hand, $R^- = (3.5 + 11 + 3.5 + 8.5) + (1.5) = 28$.
- At a confidence level of $\alpha = 0.05$ for 16 data sets, the critical value from the Wilcoxon signed-ranks test table is 29.
- Since we have 28 (which is equal or less than the critical value) as the smaller of the two sums, we reject the null hypothesis that the two classifiers are equivalent at 0.05 confidence level.

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers over Multiple Data Sets

Wilcoxon Signed-Ranks Test (cont'd):

Remark:

- Python package is available for Wilcoxon signed-ranks test: `scipy.stats: wilcoxon()`
- The Wilcoxon signed-ranks test is argued to be safer than the paired t-test since it does not assume normal distribution.
- However, when the assumptions of the paired t-test are met, the Wilcoxon signed-ranks test is less preferred over the paired t-test.

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers over Multiple Data Sets

Sign Test:

- Another popular choice, known as sign test, to compare the overall performances of classifiers is to count the number of data sets on which an algorithm is the overall winner.
- When each of the two compared algorithms wins approximately half of the data sets (i.e., $N_{win} \approx N/2$), they are assumed to be equivalent under the null hypothesis.
- The number of wins follows a binomial distribution, and the critical number of wins ($N_{critical}$) can be determined from statistical tables.
- A classifier is considered significantly better than another if it outperforms it on at least $N_{critical}$ data sets.

Ref [1]: Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research, 7:1–30.

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers over Multiple Data Sets

Sign Test (cont'd):

Example 8.3

When we apply the counting-based sign test on the above example (Table 2) where the TER-based learning was better on 11 out of 16 data sets (counting also one of the two data sets on which the two classifiers were tied), the sign test table shows that null-hypothesis cannot be rejected at confidence level of 0.05.

Null Hypothesis Significance Test (NHST)

Comparing Two Classifiers over Multiple Data Sets

Sign Test (cont'd):

Remark:

- For the example above, consider another classifier where its winning count with respect to LSE-based learning is 12 (critical value) out of the 16 data sets but with all $d_i < 0.01\%$.
- Although the null-hypothesis that the two classifiers are equivalent is rejected, the conclusion that the two classifiers are significantly different remains questionable.
- Hence much care must be taken when testing the hypothesis for small number of data sets.

Null Hypothesis Significance Test (NHST)

Comparing Multiple Classifiers over Multiple Data Sets

The above paired tests could be applied to the case of multiple classifiers by enumerating all combinatorial pairs. An argument against such application is due to the possibility of rejecting the null-hypothesis by random chance since so many tests have been made. Under this situation, either the Analysis of Variance (ANOVA, which assumes normal distribution of data) or its non-parametric counterpart, the Friedman test can be adopted.

Friedman Test:

- It is available in Matlab (statistics toolbox) and in Python (scipy.stats).

Ref [1]: Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research, 7:1–30.

Null Hypothesis Significance Test (NHST)

Comparing Multiple Classifiers over Multiple Data Sets

Friedman Test (cont'd):

Example 8.4

Table 3: Average accuracy and ranks

		Accuracy(Rank)			
	Database name	RP-FLD	RP-LSE	RP-TER	RP-AUC
1	Shuttle-l-contr	0.9811(3.0)	0.9861(2.0)	0.9909(1.0)	0.9574(4.0)
2	BUPA-liver	0.7038(2.0)	0.6995(3.0)	0.7217(1.0)	0.6513(4.0)
3	Monks-1	0.9171(1.0)	0.8959(4.0)	0.9016(3.0)	0.9161(2.0)
4	Monks-2	0.6570(1.0)	0.6174(3.0)	0.6157(4.0)	0.6471(2.0)
5	Monks-3	0.9123(1.0)	0.8977(3.0)	0.9041(2.0)	0.8745(4.0)
6	Pima-diabetes	0.7591(2.0)	0.7073(4.0)	0.7776(1.0)	0.7549(3.0)
7	Tic-tac-toe	0.9827(1.0)	0.9814(3.0)	0.9822(2.0)	0.8660(4.0)
8	Breast-cancer-W	0.9720(3.0)	0.9695(4.0)	0.9740(1.0)	0.9725(2.0)
9	Statlog-heart	0.8324(1.5)	0.8255(4.0)	0.8318(3.0)	0.8324(1.5)
10	Credit-app	0.8671(1.0)	0.8592(4.0)	0.8667(2.0)	0.8632(3.0)
11	Votes	0.9414(3.0)	0.9408(4.0)	0.9424(2.0)	0.9439(1.0)
12	mushroom	1.0000(2.0)	1.0000(2.0)	1.0000(2.0)	0.9945(4.0)
13	wdbc	0.9654(2.0)	0.9445(4.0)	0.9650(3.0)	0.9716(1.0)
14	wpbc	0.6722(2.0)	0.6218(4.0)	0.7961(1.0)	0.6591(3.0)
15	ionosphere	0.8869(1.0)	0.8372(4.0)	0.8695(2.0)	0.8663(3.0)
16	sonar	0.8162(2.0)	0.8071(4.0)	0.8152(3.0)	0.8238(1.0)
	Average	0.8667(1.781)	0.8494(3.500)	0.8722(2.063)	0.8497(2.656)

Null Hypothesis Significance Test (NHST)

Comparing Multiple Classifiers over Multiple Data Sets

Friedman Test (cont'd):

- If the null-hypothesis is rejected,a post-hoc Nemenyi test can be conducted to analyze the difference among the classifiers.
- If the average ranks of two classifiers differ by at least the *critical difference* value given by

$$CD = q_\alpha \sqrt{\frac{K(K+1)}{6N}}, \tag{3}$$

then the performance of two classifiers is significantly different.

Null Hypothesis Significance Test (NHST)

Comparing Multiple Classifiers over Multiple Data Sets

Friedman Test (cont'd):

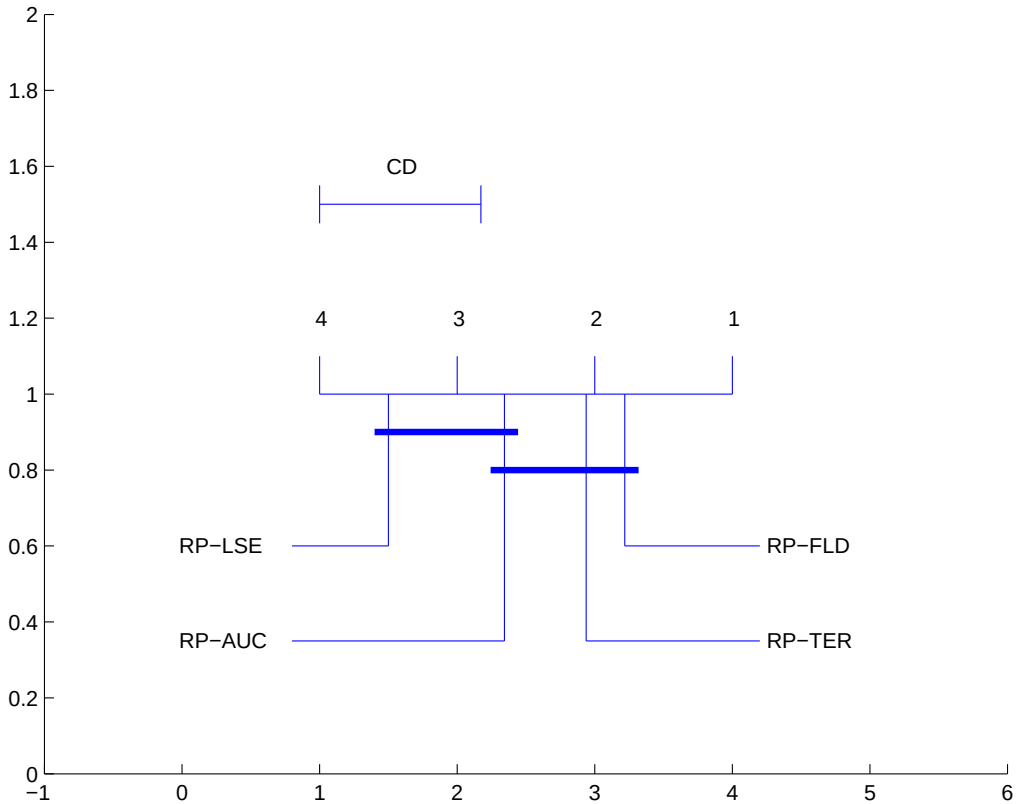


Figure 1: Comparison of classifiers against each other with the Nemenyi test where groups of classifiers that are not significantly different are connected.

Bayesian Analysis (BA)

Motivation

- Having the tools outlined for statistical inferences based on the Null Hypothesis Significance Test (NHST) in the previous section, we note here that the use of the p-value needs extra care (Ioannidis, 2005; Amrhein et al., 2019; Nuzzo, 2014).
- According to references below, the NHST has not been utilized appropriately in many instances.
- This might have caused the high rate of non-replication of research discoveries.
- In a nutshell, the two fundamental issues pointed out in these references are namely,
  - (i) the bias and the extent of repeated independent testing,
  - (ii) the use of the p-value in a dichotomous way.

[2] Ioannidis, J. P. A. (2005). Why most published research findings are false. PLoS Medicine, 2(8, e124):0696–0701.
[3] Nuzzo, R. (2014). Statistical errors: P values, the 'gold standard' of statistical validity, are not as reliable as many scientists assume. Nature, 506:150–152.
[4] Amrhein, V., Greenland, S., McShane, B., and more than 800 signatories (2019). Retire statistical significance. Nature, 567:305–307.

Bayesian Analysis (BA)

Comparing Two Classifiers on a Single Data Set

Example 8.5

Consider two classifiers namely A and B, which have respectively the following accuracies in percentage for 10 folds:

$$A: \begin{bmatrix} 95.8707 \\ 94.7935 \\ 93.5368 \\ 96.4093 \\ 95.6912 \\ 94.2549 \\ 94.6140 \\ 95.5117 \\ 96.2298 \\ 95.1526 \end{bmatrix} \quad B: \begin{bmatrix} 93.5368 \\ 92.8187 \\ 91.5619 \\ 93.8959 \\ 93.1777 \\ 93.1777 \\ 94.7935 \\ 91.7415 \\ 94.6140 \\ 94.0754 \end{bmatrix} \tag{4}$$

Bayesian Analysis (BA)

Comparing Two Classifiers on a Single Data Set

Example 8.5 (cont'd)

Then, running of Bayesian Analysis based on
`[probs,plot_data] = two_classifiers_one_dataset(x, rope, cvruns)`
(https://github.com/LucyKuncheva/Bayesian-Analysis-for-Comparing-Classifiers)
gives the posterior results: `probs = [0.0001, 0.0549, 0.9450]` where probs := $[A < B,\ A = B(rope),\ A > B]$.

Bayesian Analysis (BA)

Comparing Two Classifiers on a Single Data Set

Example 8.5 (cont'd)

The distribution of `plot_data` is plotted in Fig. 2 where the *robe* has been defined as ±0.01.

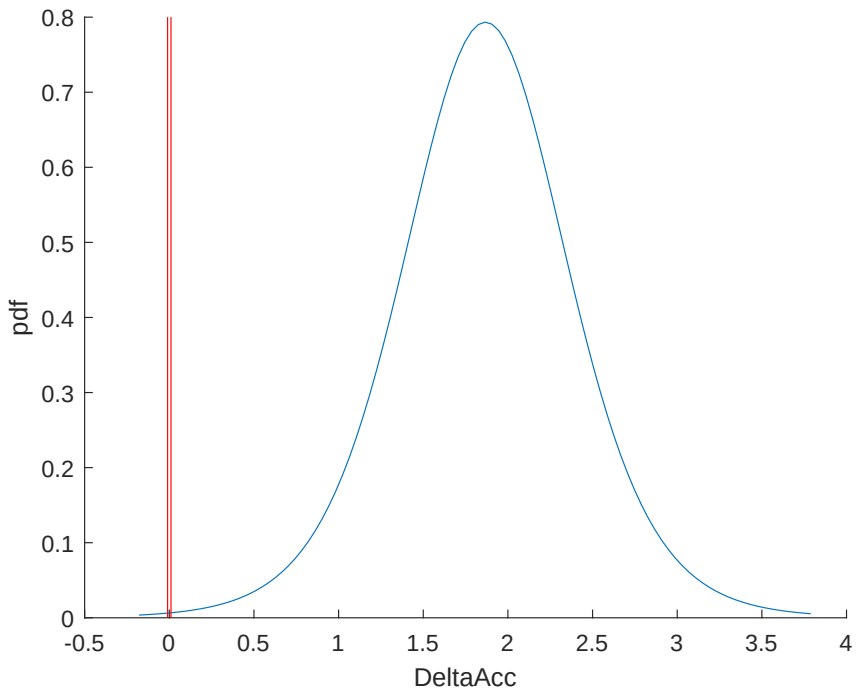


Figure 2: Posterior for A versus B in one dataset.

Bayesian Analysis (BA)

Comparing Two Classifiers on Multiple Data Sets

Example 8.6

Consider the 16 data sets of Table 2 where we want to compare between the TER method and the LSE method in term of their classification performances. Running of the following codes
`[samples, probs] = signed_rank_test_diff(x,rope,prior,T)`
(https://github.com/LucyKuncheva/Bayesian-Analysis-for-Comparing-Classifiers) gives the plot in Fig. 3. Due to the small differences in the accuracy percentage, the analysis shows a very narrow distribution with $p(\text{Rope}) = 99.834\%$, $p(\text{TER}) = 0.166\%$ and $p(\text{LSE}) = 0.000\%$.

Bayesian Analysis (BA)

Comparing Two Classifiers on Multiple Data Sets

Example 8.6 (cont'd)

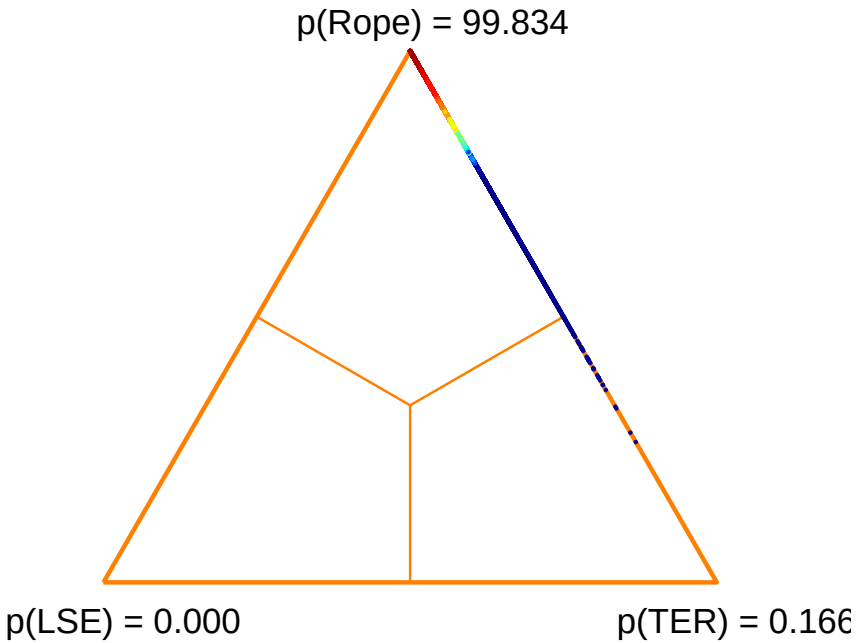


Figure 3: Bayesian analysis comparing two classifiers (TER and LSE) on the 16 data sets of Table 2.

Bayesian Analysis (BA)

Comparing Two Classifiers on Multiple Data Sets

Example 8.7

As another example, if the differences in the accuracy percentage have been enlarged by 10 times, then the analysis results in Fig. 4 is observed. This modified example demonstrates sensitiveness of the Bayesian Analysis to the effect size of differences in the accuracy percentage.

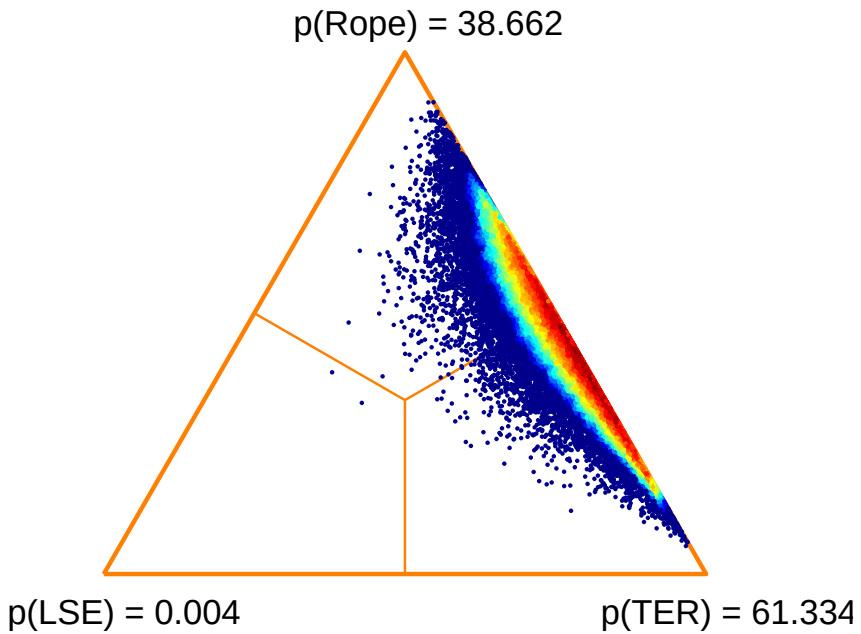


Figure 4: Bayesian analysis comparing two classifiers (TER and LSE) on the 16 data sets of Table 2 with a 10× magnification of the difference values of the classifiers' accuracies.

Bayesian Analysis (BA)

Comparing Two Classifiers on Multiple Data Sets

- According to [5] who quoted [6] that "in Bayesian analysis we usually do not have to worry about multiple comparisons.
- The reason is that we do not worry about Type I error, because the null hypothesis is hardly believable to be true."
- They suggested using multilevel analysis or hierarchical Bayesian model on multiple classifiers. In other words, the various combinations of classifier pairs are to be evaluated.

[5] Benavoli, A., Corani, G., Demšar, J., and Zaffalon, M. (2017). Time for a change: a tutorial for comparing multiple classifiers through bayesian analysis. Journal of Machine Learning Research, 18:1–36.
[6] Gelman, A., Hill, J., and Yajima, M. (2012). Why we (usually) don't have to worry about multiple comparisons. Journal of Research on Educational Effectiveness, 5(2):189–211.

Summary Notes

- While acknowledging the tendency toward a 'black-and-white' nature in null hypothesis significance testing (NHST), it is crucial to exercise caution, especially when presenting inference results and drawing conclusions.
- On the other hand, the argument in favor of Bayesian analysis (BA) that 'point-wise null hypotheses are practically always false' (based on the assumption of the existence of H_0) can be intricate.
- This complexity arises because the null hypothesis may not apply equally to two groups with diverse observations drawn from the same classifier.
- Given the current available tools, we recommend using BA for comparing two classifiers on a single data set or across multiple data sets.
- Additionally, consider reserving the post-hoc Nemenyi test for visualizing the ranking order of multiple classifiers. Most importantly, avoiding absolute 'black-and-white' claims can prevent misrepresentation.

Thank you!

Amrhein, V., Greenland, S., McShane, B., and more than 800 signatories (2019). Retire statistical significance. *Nature*, 567:305–307.

Ioannidis, J. P. A. (2005). Why most published research findings are false. *PLoS Medicine*, 2(8, e124):0696–0701.

Nuzzo, R. (2014). Statistical errors: *P* values, the 'gold standard' of statistical validity, are not as reliable as many scientists assume. *Nature*, 506:150–152.